

SIMATS ENGINEERING

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36. Polyphase Decimator for $M = 4$

Inefficiency

For a polyphase decimator with $M = 4$, the FIR filter is divided into four polyphase branches. The implementation is inefficient if all four branches are executed unnecessarily for every output sample. This causes redundant multiplications and additions and increases the computational load.

Cause

The main cause is incorrect phase and index handling. The decimator should use the four filter phases according to the decimation factor. If the branch calculations are repeated instead of being selected and organized correctly, the computational advantage of polyphase decomposition is lost.

Correction

The filter coefficients should be separated into four phases:

- Phase 0: $h[0], h[4], h[8], \dots$
- Phase 1: $h[1], h[5], h[9], \dots$
- Phase 2: $h[2], h[6], h[10], \dots$
- Phase 3: $h[3], h[7], h[11], \dots$

The indexing and control logic must be corrected so that duplicate branch operations are avoided and only the required filtering operations are used to generate the decimated output.

Computational Savings

Let the original FIR filter have N coefficients. With $M = 4$, each of the four polyphase branches contains approximately $N/4$ coefficients. If all four branches are unnecessarily evaluated, about N multiplications are performed per output sample. After proper correction, approximately $N/4$ multiplications are required for the branch-selective case.

Computational saving = $N - N/4 = 3N/4 = 75\%$

For example, if the filter has 40 taps, the four branches contain about 10 taps each. Unnecessarily evaluating all four branches requires about 40 multiplications, whereas the corrected branch-selective operation requires about 10. Hence, about 30 multiplications are saved per output sample.

Application

Polyphase decomposition reduces unnecessary computation in multirate digital signal processing. For a decimation factor of 4, the output sampling rate is one-fourth of the input sampling rate. Splitting the filter into four phases allows the filtering and downsampling operations to be implemented more efficiently.

Conclusion

The inefficiency is due to redundant execution of polyphase branch calculations. Correct phase selection and indexing remove the unnecessary operations. For the branch-selective implementation considered here, the estimated computational saving is approximately 75%.